String separators, normal modules and transitive rank

A supplement to the complete Topic B common theory

12 October 2026

Let G be a finite group generated by involutions $r_0, \dots, r_{d-1}$, all distinct and nonidentity. It is a string C-group if $[r_i, r_j] = 1$ when $|i - j| > 1$, and with the intersection property $\langle r_i : i \in I\rangle \cap \langle r_j : j \in J\rangle = \langle r_i : i \in I \cap J\rangle$. In particular its generators, and every subfamily, are independent: no generator belongs to the subgroup on the others. Its noncommuting diagram is the subgraph of the labelled path having an edge precisely when the adjacent generators fail to commute. An interval of labels need not be a connected component of this graph.

This supplement uses the complete accepted Topic B common manuscript *Forest coordinates and relative generation*, not an unpaid application theorem. Its full sixteen pages remain an explicit common input. We use its normal-quotient and normal-module bounds, extremal-tree theorem, and the weak primitive independent-length bound. The latter is needed for block images, which need not inherit IP. We do not use the older general imprimitive irredundant-length theorem: the normalized two-block image does inherit full IP, and strong induction on the faithful permutation degree supplies its sharper bound. Whiston's bound and equality clause, original Maróti classification, and the classical identifications stated there remain retained premises. The separate Conder lower-order theorem is used only in the sharp primitive section and is explicitly retained there.

The block normalization is borrowed from [CFLM14]. The new proof supplies the two-neighbor case omitted from their two-large-block argument, a complete finite residual check, and an explicit normal-module and cocycle treatment of equality. It does not claim priority for their rank theorem or for the borrowed normalization. Ordered string tuples and unordered marked sets are kept distinct.

1 Primitive quotients on a maximal block system

Let a transitive string C-group on km letters preserve a maximal nontrivial block system with m blocks of size k. Its block image P is primitive. Choose an inclusion-minimal distinguished subfamily L generating P, and put C equal to the distinguished generators outside L commuting with every member of L. Denote the remaining family by R.

If the noncommuting diagram of the images of L is disconnected, its commuting factors are normal in P. A primitive nontrivial normal subgroup is transitive; two commuting transitive subgroups are regular. Their intersection is central and trivial. Each regular normal factor must be simple, since a proper characteristic or normal subgroup would give a block system for the product. Thus $P = T \times T$ in the left–right action on a nonabelian simple group T, and $m = |T| \geq 60$. There are exactly two diagram components. Each irredundant generating family has length at most $\log_2 |T|$, so $|L| \leq 2 \log_2 m$. There are at most four adjacent positions in R.

Otherwise the images of L have a connected diagram. Therefore the corresponding distinguished vertices occupy an interval and $|R| \leq 2$. Whiston gives $|L| \leq m - 1$, with equality only for $P = S_m$. The primitive block action is not cyclic of odd prime order, since it is generated by involutions. Except when $m = 2$, it consequently has trivial centralizer. Thus $\langle C\rangle$ fixes each block. Its restriction to one block is faithful: an element trivial on that block and commuting with $\langle L\rangle$ is trivial on all blocks, since L is transitive on them. Hence $|C| \leq k - 1$.

For the disconnected case this gives

$$d \le 2\log_2 m + k + 3 < km/2 \qquad (m \ge 60,\ k \ge 2).$$

For example the difference $km/2 - k - 3 - 2\log_2 m$ is increasing in k and $m \ge 60$, and is positive at $(k, m) = (2, 60)$.

Suppose now $k, m > 2$ and L is connected. Initially $d \le k + m$. Equality would make $P = S_m$ and $\langle C \rangle$ a diagonal S_k on the blocks. The diagram of C is connected because S_k is centerless and directly indecomposable. Both neighbors of the L interval must be present; all of C lies on one side of it. The opposite neighbor centralizes C. In the full block stabilizer, the centralizer of this diagonal S_k is precisely the group S_m permuting the blocks consistently. Since L lies in this centralizer and maps onto S_m, it is the whole centralizer. The opposite neighbor is therefore already in $\langle L \rangle$, contrary to independence. Consequently

$$d \le k + m - 1 \le km/2 \quad (k, m \ge 3,\ km \ge 10).$$

The last inequality follows from $(k-2)(m-2) \ge 2$; the only excluded pair is $(3, 3)$, which is part of the finite degree-nine boundary.

2 Two-point blocks and the equality marking

Assume $k = 2$ and $m \ge 5$. Write $V = \mathbb{F}_2^m$ for the base module of $C_2 \wr S_m$, $W = G \cap V$, and $P = G/W$. Any member of C is the constant flip $\lambda = (1, \ldots, 1)$, so $|C| \le 1$. For a connected L we have $|R| \le 2$. If $P \ne S_m, A_m$, the accepted primitive independent-length bound $\mu'(P) \le m-3$ gives $d \le m$, apart from the primitive degree-six copy of S_5.

The A_m-invariant subspaces of V, for $m \ge 5$, are $0, \langle\lambda\rangle, E, V$, where E is the even-weight space (with the usual coincident or nested spaces understood). Indeed, from any nonconstant vector a three-cycle gives a nonzero difference of weight two. Conjugating by A_m, which is transitive on unordered pairs, generates E. A family independent under generation by the group has at most two members, so the normal-module inequality and $\mu(A_m) \le m - 2$ give $d \le m$.

The exceptional degree-six $S_5 = PGL_2(5)$ is three-transitive. Every nonconstant vector of weight one, two or three (or its complement) has conjugate differences generating E; for weight three use three-transitivity, and for the other cases two-transitivity suffices. Thus it has the same subspace list. Since $m = 6$ is even, every invariant subspace has module-independent length at most one: an odd vector generates V, a nonconstant even vector generates E, and a constant vector generates its line. The normal-module inequality gives $d \le \mu(S_5) + 1 \le 5 < 6$.

It remains to consider $P = S_m$. The same module list gives $\mu'_G(W) \le 2$, so $d \le m + 1$. When m is even the preceding one-generator argument improves this to $d \le m$. Suppose m is odd and $d = m + 1$. Equality in the normal-quotient proof forces $|L| = m - 1$. Its block images form a Coxeter path: this is UB.Extremal for $m \ge 7$, and the complete independent-string certificate for $m = 5, 6$. They occupy an interval in the full string. Each of the two outside generators centralizes the block subgroup omitting at most one endpoint of that path. Its centralizer in S_m is trivial (also for the outer S_6 image), so both outside generators actually belong to V.

A vector commuting with every adjacent block transposition except possibly the first or last is constant off that exceptional endpoint. The possible nonzero vectors are therefore $\lambda, e_i, \lambda + e_i$, where i is an endpoint of the block path. The vector e_i generates all of V under S_m; two nonconstant even vectors each generate E. Independence of both outside generators consequently forces them to be λ and $\lambda + e_i$. In particular $W = V$. The subgroup $\langle L \rangle$ has zero intersection with V: any nonzero S_m-submodule there contains either λ or E, making one of these two outside generators redundant. Thus $\langle L \rangle$ is a complement, the graph of a 1-cocycle $c : S_m \to V$. The automorphism

$$(v, \sigma) \longmapsto (v + c(\sigma), \sigma)$$

of $V \rtimes S_m$ fixes the base module and sends this complement to the standard one. As an unordered marked set, up to a group isomorphism carrying the set of distinguished generators onto the other set, the marking is exactly

$$\lambda, \quad \lambda + e_1, \quad s_1, \ldots, s_{m-1}, \qquad s_i = (i, i+1).$$

Its displayed ordered Schlaefli type is $[2, 4, 3, \ldots, 3]$. The noncentral path may be ordered in either direction, and the isolated central generator may be placed at either end. These are the four compatible string orders, forming two classes under global duality, with representatives $[2, 4, 3, \ldots, 3]$ and $[2, 3, \ldots, 3, 4]$. We do not identify these two ordered polytopes: their labelled types differ and neither is the global reverse of the other. They share the same unordered marked set. In particular the source's page-two type belongs to the reversed noncentral path, not to its literal displayed tuple; its page-seven type matches that tuple. The source's uniqueness claim for the unordered pair (G, S) is retained with this distinction.

For completeness the marking has IP for every odd $m \geq 3$. The central factor $\langle \lambda \rangle$ is disjoint from $E \rtimes S_m$. In the remaining family $v = \lambda + e_1, s_1, \ldots, s_{m-1}$, a subset of adjacent transpositions generates the Young subgroup P_T on its interval partition. If v is absent the base part is zero. If v is present and the initial interval containing 1 is A, the base part is

$$W_A = \operatorname{span}\{\lambda + e_i : i \in A\}.$$

For proper A the indicated vectors are independent, since their common outside coordinate determines the sum of their coefficients. For $A = [m]$ their only relation is the sum of all m vectors, giving $W_{[m]} = E$. All initial intervals are nested and therefore $W_A \cap W_B = W_{A \cap B}$. The corresponding Young subgroups satisfy $P_T \cap P_U = P_{T \cap U}$, because their interval partitions refine by the intersection of their edge sets. The actual subgroups are $W_A \rtimes P_T$, so their intersections satisfy IP. Adjoining the independent central factor preserves it. The full group is transitive on the $2m$ signed points and is $C_2 \wr S_m$.

3 Two general equality lemmas

Lemma 1. *Let $H \leq S_k$ be intransitive and generated irredundantly by $k - 2$ elements. Then $H = S_A \times S_B$ on exactly two nonempty orbits A, B partitioning the k letters.*

Proof. Successive orbit projections and the normal-quotient normalization bound give independent length at most $\sum_O (|O| - 1) = k - q$, where q is the number of orbits. Hence $q = 2$. Write $a = |A|$, $b = |B|$. Choose a minimal subfamily T generating the first projection. It has at most $a - 1$ members. Normalize all other generators into the second-coordinate kernel by words in T. They remain independent, so there are at most $b - 1$ of them. Equality forces both bounds to be attained. Whiston's equality clause makes the first image S_A and the subgroup generated by the normalized extra elements S_B. Thus the whole second factor is in H, and the full first image then gives $H = S_A \times S_B$. An orbit of size one has the trivial symmetric factor and contributes zero throughout. □

We use the following extremal consequence of UB.Extremal: an independent string marking of S_b with $b - 1$ involutions, for $b \geq 7$, is a Coxeter path. Indeed a sign-modified tree marking cannot consist entirely of involutions: an edge adjacent to the distinguished edge gives an element of order three. For the ordinary tree, the string condition makes its line graph a path, so the tree itself is a path. For $b = 5, 6$ the complete small certificate verifies the same abstract Coxeter-path conclusion, including the outer-automorphism image in S_6. It also verifies the trivial centralizer of the subgroup omitting an endpoint.

Lemma 2. *For $a = 2$ or 3, and $b \geq 6$, an involutory string C-marking of $S_a \times S_b$ of rank $a + b - 2$ has two nonempty diagram components. One generates S_a, and the other is the graph of a homomorphism $S_b \to S_a$; this homomorphism is trivial when $a = 3$.*

Proof. Choose a minimal subfamily L generating the S_b quotient. Normal-quotient normalization bounds the remaining family by $a-1$, so $|L| = b-1$. The images of L are an extremal Coxeter path. Its noncommuting adjacent vertices must be adjacent in the full string, so L occupies an interval. Every generator outside that interval centralizes all but possibly one endpoint of this path. The endpoint-omission subgroup has trivial centralizer in S_b; therefore every outside generator has trivial S_b image and lies in the small factor. There are exactly $a-1$ of them.

Put $N = \langle L\rangle \cap S_a$. Their irredundance relative to $\langle L\rangle$ implies irredundance relative to N under conjugation by $\langle L\rangle$. If $a = 2$, the one extra generator forces $N = 1$ and is central; its diagram vertex is isolated, and $\langle L\rangle$ is a graph over S_b. If $a = 3$, any nontrivial subgroup of S_3, together with a family generating S_3, needs at most one member of that family. Thus two independent extras force $N = 1$. The complement $\langle L\rangle$ is the graph of a homomorphism $S_b \to S_3$, whose image is trivial or order two, since A_b is simple and $b \geq 5$. If the image has order two, one of the two extra involutions lies outside it. That involution and its conjugate by the image generate S_3, making the other extra redundant. Hence the image is trivial. Now the two small-factor involutions commute with all of L, so the diagram is disconnected. □

4 The uniform two-large-block obstruction

Let a transitive string C-group G preserve two blocks of size k. Choose a distinguished generator r interchanging the blocks, nearest to an endpoint of the string among all such generators. Identify the blocks through r, so r is the pure block swap in $S_k \wr C_2$. Let C be the other distinguished generators commuting with r, and R the remainder. Only its string neighbors can be in R, so $|R| \leq 2$.

The centralizer of r in this wreath product is $\mathrm{diag}(S_k) \times \langle r\rangle$. Multiplying each $c \in C$ by r when necessary normalizes it into the diagonal copy of S_k. The projection is injective on $\langle C\rangle$, because the intersection property gives $r \notin \langle C\rangle$. Its image H therefore has the same involutory string C-marking. Moreover

$$\langle C\rangle = \{(h,h)r^{\chi(h)} : h \in H\}$$

for a character $\chi : H \to C_2$. Whiston gives $|C| \leq k-1$, and consequently $d \leq k+2$.

Theorem 1. *For $k \geq 9$, suppose every transitive string C-group of faithful degree k, other than S_k, A_k, has rank at most $k/2+1$. Then a transitive two-block string C-group on $2k$ letters has $d \leq k$.*

Proof. If $d = k+2$, then $|C| = k-1$ and $H = S_k$. Its diagram is a Coxeter path, so C lies on one side of r and its neighbors: r must be one place from a string endpoint. The outside neighbor centralizes all of C. But $\ker\chi$ contains A_k, whose centralizer in S_k is trivial. An element $(p,q)r^{\epsilon}$ centralizing C therefore has $p = q = 1$, and is in $\langle r\rangle$, a contradiction.

Suppose $d = k+1$. If r is at an endpoint, $|C| = k-1$ and the one remaining generator centralizes a Coxeter endpoint-omission subgroup of C. Its character kernel contains an A_{k-1} fixing one letter; that subgroup again has trivial centralizer in S_k. The remaining generator is in $\langle r\rangle$, a contradiction. If r is interior but $|R| = 1$, then again $|C| = k-1$ and its diagram is connected. Therefore r is one place from an endpoint and the sole noncommuting neighbor is on that outer side. It commutes with all of C, so the preceding A_k centralizer argument excludes it. Thus $|R| = 2$ and $|C| = k-2$.

First let r have distance one from an endpoint, so C occupies one interval (its noncommuting diagram need not be connected). Its outside neighbor centralizes all of C and does not commute with r. If H is transitive, the hypothesis at degree k and $k-2 > k/2+1$ leave only S_k, A_k. This use is legitimate because the projection onto H preserves the full marking and IP. The same character-kernel centralizer argument excludes both. Otherwise Lemma 1 gives $H = S_A \times S_B$. If both orbits have size at least four, $\ker\chi$ contains $A_A \times A_B$ and has trivial centralizer. A singleton orbit is also excluded by this argument on its complement. The only remaining possibilities have

$a = |A| \in \{2, 3\}$ and $b = |B| \geq 6$. If χ is nontrivial on the large factor, a centralizing $(p, q)r^\epsilon$ first has p, q supported on A (by A_B), then has $p = q$ by an odd permutation supported on B. It commutes with r, a contradiction. If χ is trivial there, Lemma 2 gives two components. The component at the far endpoint has extremal Coxeter generators, all odd in its symmetric factor. A nontrivial restriction of χ to that component would make its endpoint a block swap closer to an endpoint than r. Hence χ is trivial on that far component. Both neighbors of r centralize it: the inner neighbor commutes with all of C except possibly its first vertex, in the other component. The far component has A or B as its distinguished orbit; any other nontrivial orbit has a different size. Its centralizer in S_k therefore preserves A, B. Thus the coordinate permutations of both neighbors preserve A, B. Together with C and r they preserve $A \times \{0, 1\}$, contradicting transitivity.

Finally let r have distance at least two from both endpoints. Then C occupies two nonempty intervals separated by the missing neighbors and r. Their generated images commute and form a direct product decomposition of H, by the intersection property; we do not yet assume that either interval has connected noncommuting diagram. The same smaller-degree hypothesis excludes transitive H other than S_k, A_k, and these cannot be a nontrivial direct product. Thus again $H = S_A \times S_B$ by Lemma 1. Neither orbit is a singleton. For orbit sizes at least three, the symmetric factors are centerless and directly indecomposable, so the two interval factors are precisely those symmetric factors. Their noncommuting diagrams are therefore connected. If one orbit has size two, one component is this central C_2, and the other is a graph of a character $S_B \to C_2$. This graph is directly indecomposable too. Thus in both cases the intervals are exactly the two noncommuting components, and each attains the maximum independent length in its symmetric factor.

If χ is nontrivial on such a component of size at least five, its extremal Coxeter generators are all odd. Hence the endpoint of that component at the end of the whole string interchanges the two large blocks, contradicting the choice of r. The same holds for factors of size two or three. For S_4, at least one of the first two and one of the last two generators is odd: otherwise two even involutions lie in its abelian normal four-group, and the string relations make an endpoint generator central in S_4. This again gives a closer block-swapping generator. Therefore χ is trivial on both components, and all of C fixes both large blocks individually.

Each member of R centralizes the opposite component of C. Its coordinate permutations consequently preserve the two sets A, B. This remains true for the possible character graph over the large symmetric factor when $|A| = 2$: its two nontrivial orbits have different sizes. All of C, R and r therefore preserve $A \times \{0, 1\}$ and $B \times \{0, 1\}$, contradicting transitivity. □

5 The finite two-block boundary

The same argument excludes $d \geq k + 2$ for $5 \leq k \leq 8$. For $k = 5, 6$ the complete independent-string certificate proves both that the extremal markings are Coxeter paths and that the derived subgroup of an endpoint-omission subgroup has trivial centralizer. In the outer S_6 marking that subgroup is a transitive degree-six A_5, not a point-fixing alternating group. These actual centralizer checks replace the natural point-fixing description in the endpoint argument. The cases with $d = k + 1$ and $|R| \leq 1$ are excluded in the same way.

For $d = k + 1$ and $|R| = 2$, the faithful normalized $H \leq S_k$ has rank $k - 2$. If it is intransitive, Lemma 1 gives $H = S_a \times S_b$. A singleton orbit is excluded by the alternating centralizer argument, or by direct indecomposability when the two intervals are nonempty. The remaining unordered orbit sizes are $(2, 3), (2, 4), (3, 3), (2, 5), (3, 4), (2, 6), (3, 5), (4, 4)$. For a transitive H, the accepted primitive bounds and the complete small transitive-imprimitive census leave only S_k, A_k, the primitive degree-six S_5, and the degree-six actions numbered 9 and 11. The first two are excluded as above. In the degree-six S_5 case, its derived subgroup is primitive A_5 with trivial centralizer, so the same argument applies. The degree-eight census has no rank-six marking; degree five and seven have no nontrivial imprimitive actions.

For the eight product actions and the two degree-six actions, the separate, completely proved finite-neighbor lemma enumerates every C-marking, character, compatible split and pair of missing neighbors. Its 59 normalized markings yield 154 transitive completions, all redundant. This exact required finite premise therefore excludes all remaining cases. Consequently the two-large-block bound $d \le k$ holds for every $k \ge 5$. The block sizes three and four are included in the separately complete degree-six and degree-eight census; they contain genuine exceptional markings.

6 A sharp primitive bound with a bounded residual census

We now assume that G is primitive of degree n, distinct from A_n, S_n. If its noncommuting diagram is disconnected, the earlier primitive commuting-factor argument gives $G = T \times T$ on $|T| = n \ge 60$ points and $d \le 2\log_2 n < n/2$. Otherwise every adjacent pair fails to commute. We explicitly retain the following independent theorem of Conder: a finite nondegenerate regular polytope of rank $r \ge 9$ has at least 2^{2r-1} flags, equivalently its string C-group has at least that order. We use only this stated rank range and only connected string diagrams. This is an external theorem, not a consequence of our finite diagnostics.

Apply Maróti's original primitive-order theorem, including its lower containment $A_m^\ell \le G \le S_m \wr S_\ell$ in the product-action alternative, with $m \ge 5$ and, after replacing subsets by complements, $1 \le k \le m/2$. The degree is $n = \binom{m}{k}^\ell$. If $\ell = 1$, the natural $k = 1$ action would give the excluded A_n, S_n. For $k \ge 2$, Whiston in the faithful m-point action gives $d \le m - 1 < n/2$. If $\ell \ge 2$, the same abstract group has a faithful transitive imprimitive action on $m\ell$ letters: the lower subgroup A_m^ℓ is transitive in each block and primitivity of the product action makes the coordinate action transitive. The induction hypothesis at the strictly smaller degree $m\ell < n$ provides the imprimitive bound and gives $d \le m\ell/2 + 1 < n/2$, since $m\ell + 2 < m^\ell \le \binom{m}{k}^\ell$ for $m \ge 5, \ell \ge 2$.

The Mathieu alternative is treated by the bounded checks below. In the remaining alternative put

$$M(n) = n \prod_{i=0}^{\lfloor \log_2 n \rfloor - 1} (n - 2^i).$$

Exact integer arithmetic verifies $M(n) < 2^{n-1}$ for $26 \le n \le 63$. For $n \ge 64$, with $q = \lfloor \log_2 n \rfloor \ge 6$, we have $M(n) < n^{q+1} < 2^{(q+1)^2} \le 2^{n-1}$; the inequality $(q+1)^2 \le 2^q - 1$ holds at $q = 6$ and persists by induction. If $d \ge n/2$, then $d \ge 13$, so Conder gives $|G| \ge 2^{2d-1} \ge 2^{n-1} > M(n)$, a contradiction. We use the exact product bound here, not the source's coarser logarithmic estimate at its asserted small threshold.

6.1 Complete finite screening

For $2 \le n \le 25$, the required program visits every primitive catalogue entry, omitting exactly A_n, S_n. It uses the following necessary bounds, with all finite groups and actions computed exactly. If $\Omega(t)$ counts prime factors with multiplicity, then $d \le \Omega(|G|)$, by the strictly increasing subgroup chain from IP. All distinguished generators are involutions, so a proper subgroup generated by all involutions excludes G outright. For every possible endpoint x, the subgroup on labels $2, \dots, d-1$ commutes with x and intersects $\langle x \rangle$ trivially. Consequently

$$d \le 1 + \Omega(|C_G(x)|).$$

Likewise the even labels generate an elementary abelian subgroup of rank $\lceil d/2 \rceil$. If every elementary abelian subgroup of a Sylow two-subgroup has rank at most e, then $d \le 2e$. The finite program computes this bound by the complete subgroup list of that Sylow group; no assertion about its mere minimum number of generators is substituted for its elementary abelian rank.

For primitive affine degree sixteen, the program verifies the regular normal subgroup 2^4. The normal-module lemma costs one generator, and the faithful quotient embeds in $GL_4(2) \cong A_8$, whose independent families have length at most six by Whiston. Thus $d \leq 7 < 8$. The exceptional isomorphism is explicitly checked and is also the retained identification already used by common Topic B. For the degree-nine affine group $AGL_2(3)$, the program verifies $\Phi(GL_2(3)) \cong C_2$ and the Frattini quotient S_4; its maximum irredundant generating length is at most three. The irreducible translation module costs one, giving $d \leq 4 < 9/2$. The degree-fifteen copy of A_8 has $d \leq 6$ directly.

For both actions of M_{11}, the Sylow elementary abelian rank is two, so $d \leq 4$. The degree-twenty-three group M_{23} is already excluded by the endpoint-centralizer bound. For M_{24} the two involution centralizers have orders $7680, 21504$. If $d \geq 12$, the connected tail on labels $2, \ldots, d-1$ has rank at least ten. Conder applied to that tail, disjoint from $\langle x \rangle$, requires $|C_G(x)| \geq 2 \cdot 2^{2(d-2)-1} = 2^{2d-4} \geq 2^{20}$, contradicting both actual centralizer orders.

The surviving actions are exactly the following GAP entries:

$$(6,1), (6,2), (7,5), (8,3), (8,5), (10,4), (10,5), (10,7), (12,2).$$

The degree-eight copy $(8,4)$ is explicitly identified with $(7,5)$. These are GAP's current indices, not Sims' indices printed in the source. The maximum ranks searched are respectively $3,4,4,5,4,6,6,6,6$. For A_5, S_5 they follow from Whiston; for $PSL_3(2)$ and $AGL_3(2)$ from the accepted common independent-length bounds; for the others from the checked Sylow elementary abelian ranks two or three.

We enumerate connected involutory strings up to simultaneous normalizer conjugation. At each prefix we discard dependence, failed string commutations, or a failed endpoint-parabolic intersection. These are necessary conditions for every subfamily of a C-marking. If H is the prefix subgroup and at least t future generators remain after the next one, their group centralizes H and is disjoint from it; hence $[C_G(H) : Z(H)] \geq 2^t$. This necessary inequality supplies a further pruning rule. At a full generating tuple every pair of subset subgroups is tested for IP. Stabilizer orbits at each extension preserve the full prefix and cannot discard an ordered conjugacy class. No sample markings, probabilistic generation test or source rank table is used.

The complete outputs, up to duality, are

n	G	types at rank at least $n/2$
6	A_5	$[3,5],\ [5,5]$
6	S_5	$[3,3,3],\ [4,5],\ [4,6],\ [5,6],\ [6,6]$
10	S_6	$[3,3,3,3]$.

The last group is GAP $(10,5)$ and is identified exactly with S_6. The full permutation tuples are retained, including both dual orientations. Only the rank-four degree-six simplex exceeds half the permutation degree. Thus this recovers the complete primitive branch of Theorem 1.2 and the exception list of Proposition 3.2. The source's finer Table 3 maximum-rank numbers are not imported as premises of this proof; any not independently computed are separately attributed ancillary reports in the source map.

7 Closing the simultaneous induction

Theorem 2. *For every faithful transitive string C-group of degree n, other than A_n, S_n, the rank is at most $n/2+1$. In the imprimitive case with $n \geq 10$, equality requires two-point blocks with an odd number of blocks, and has exactly the markings classified above. In the primitive case the rank is strictly below $n/2$ except for the explicitly enumerated degree-six and degree-ten markings.*

Proof. We prove these assertions simultaneously by strong induction on n. The small imprimitive actions of degrees six, eight and nine are covered by the complete required census. It searches

through rank $n-2$, the Whiston bound for a proper subgroup other than S_n, without using any imprimitive rank theorem. A prime degree has no nontrivial block system. For degree four the possible imprimitive groups are subgroups of D_8; its Frattini quotient is C_2^2, so any independent generating family has at most two members. The primitive checks through degree twenty-five are independent of the desired induction assertion. These observations supply every small endpoint needed below.

At degree n, the maximal-block argument handles $k, m > 2$. For blocks of size two it uses only the previously accepted independent-length bounds on primitive quotient images, the explicit binary-module lemma, and the complete extremal S_5, S_6 checks. It does not apply a C-group rank theorem to a quotient that might have lost IP. For two blocks, the finite ten-action lemma treats $5 \le k \le 8$. When $k \ge 9$, the normalized H is itself a string C-group of degree $k < n$, so the induction hypothesis supplies precisely the assumption in Theorem 1. Thus the whole imprimitive conclusion at degree n follows with its complete equality classification.

For a primitive group, the product-action alternative of Maróti has a faithful imprimitive action of degree $m\ell < n$. The induction hypothesis, not a same-degree result, applies there. The other primitive alternatives use Whiston, the retained Conder theorem, or the independent bounded catalogue checks, as proved above. This proves the primitive conclusion at degree n. Every invocation of the asserted sharp rank bound has therefore decreased the faithful permutation degree, closing the induction without using the older general imprimitive independent-length theorem. □

A Complete retained finite-neighbor proof

The following full body is copied exactly from the root-authored bounded note, independently accepted by the separate finite-boundary reviewer. Its original wrapper is retained in the archive, and its entire mathematical content is part of this common manuscript. The complete programs appear in our appendix. We establish the finite obstruction used in the two-block argument. The classification of the possible groups in this statement is a separate obligation: no conclusion about other groups is implicit in the calculation below. In particular, this component is not a proof of the full transitive rank theorem.

Let $W = S_k \wr C_2$ act on two blocks of size k, and let r be the permutation exchanging their corresponding letters. For a subgroup $H \le S_k$ and a character $\chi : H \to C_2$, put

$$D_\chi(H) = \{(h,h)r^{\chi(h)} : h \in H\}.$$

The two symmetric factors in a row $S_a \times S_b$ of the table below act on disjoint sets of sizes a, b in one block. The two transitive degree-six actions are the literal permutation groups numbered 9 and 11 in the GAP transitive-group catalogue; their orders are 36 and 48, respectively. The action, rather than the abstract group alone, is part of the specification.

Lemma 3. *Let H be one of the ten actions in Table 1, of degree k. Take an involutory string C-marking $(h_1, \dots, h_{k-2})$ of H, and let C be its image in $D_\chi(H)$. Split this ordered list as C_L, C_R, requiring the two lists to commute elementwise. There is no tuple*

$$(C_L, x, r, y, C_R)$$

of independent involutions that generates a transitive subgroup of W, satisfies the string commutation relations, has both x and y noncommuting with r, and has r at least as close to an endpoint as every other generator exchanging the blocks.

Proof. We give the complete finite enumeration and its exhaustion argument. The accompanying program uses exact finite permutation groups.

First enumerate the involutory string C-markings of H. At a prefix $(h_1, \dots, h_j)$, an eligible next generator is an involution of H outside the prefix subgroup and commuting with $h_1, \dots, h_{j-1}$.

We discard a prefix if one of its generators belongs to the subgroup generated by the others. Every subfamily of an independent family is independent, so this discards no desired tuple. Also, if the current subgroup has index less than 2^{k-2-j} in H, the prefix cannot acquire the required number of independent generators: each strict subgroup enlargement has index at least two.

Simultaneous conjugation by $N_{S_k}(H)$ preserves every condition. At each step choose one orbit representative under the pointwise stabilizer of the prefix under conjugation, then replace that stabilizer by its centralizer of the chosen generator. Induction on the prefix length shows that every ordered marking is represented. At length $k-2$, require generation of H and test the intersection property for every pair of subsets of the marking. The test forms the actual generated subgroups and their intersections, without a quotient or an assumed presentation.

For each surviving marking enumerate all 2^{k-2} assignments of its generators to C_2. Retain exactly those assignments for which `GroupHomomorphismByImages` constructs a homomorphism. Since the marking generates H, these are exactly its characters. Construct the corresponding diagonal lifts explicitly as permutations of $2k$ letters. Try every cut between the lifts, including the two empty-end cuts, and require elementwise commutation across it. Reject a cut and character only if an already fixed block-exchanging generator is closer to an endpoint than r.

The string relations require x to centralize all of C_R and all but the last member of C_L. Similarly, y must centralize all of C_L and all but the first member of C_R. Compute these two centralizers inside W, enumerate their involutions, and impose noncommutation with r and the same endpoint restriction. We may choose one representative for x under the simultaneous centralizer of C and r in W: conjugation by this group preserves the complete candidate set for y. For each remaining pair require $xy = yx$ and transitivity. Finally test independence by deleting each generator in turn and comparing the resulting subgroup with the full group.

The exhaustive counts are in Table 1. No transitive completion occurs in the first eight cases. In the last two cases, all transitive completions fail independence. Thus no test of the full intersection property is needed to reject a completion, proving the lemma. □

Action of H	k	Markings	Transitive completions	Independent
$S_2 \times S_3$	5	4	0	0
$S_2 \times S_4$	6	12	0	0
$S_3 \times S_3$	6	1	0	0
$S_2 \times S_5$	7	4	0	0
$S_3 \times S_4$	7	6	0	0
$S_2 \times S_6$	8	8	0	0
$S_3 \times S_5$	8	2	0	0
$S_4 \times S_4$	8	9	0	0
TransitiveGroup$(6, 9)$	6	1	2	0
TransitiveGroup$(6, 11)$	6	12	152	0

Table 1: Complete bounded search. Markings are represented up to the normalizer conjugations described in the proof. Completion counts use the stated pruning of x; they are not counts of isomorphism classes.

The implementation is in `search.g`; `run.g` executes all ten cases, and `check.py` requires their exact summaries and successful completion. To check that the procedure can retain valid completions, it is also run with $k = 3$ and the empty marking of the trivial group. It recovers six independent transitive rank-three string C-markings after the same pruning. For example,

$$((5\ 6),\ (1\ 4)(2\ 5)(3\ 6),\ (1\ 2))$$

passes the full intersection test. A redundant triple of distinct transpositions in S_3 supplies negative controls for independence and the intersection property.

The proof depends on GAP's exact finite permutation-group algorithms and its degree-six transitive catalogue. It does not supply a Lean formalization. The separate reduction of an arbitrary two-block group to these ten actions, the endpoint cases, and all other branches of the transitive rank theorem remain outside this lemma. Normalizing the commuting generators by powers of the block swap already appears in Cameron, Fernandes, Leemans and Mixer, *String C-groups as transitive subgroups of* S_n, arXiv:1410.5863v1 (2014), proof of Proposition 2.1, Case 3. We retain that step. The present contribution is the explicit character description and exhaustive finite analysis with two noncommuting neighbors, within the common string-separator argument. It neither claims a new open-conjecture solution nor cites the target rank theorem as a premise.

References

[1] P. J. Cameron, M. E. Fernandes, D. Leemans and M. Mixer, *String C-groups as transitive subgroups of* S_n, arXiv:1410.5863v1, 21 October 2014, twelve-page edition.

[2] P. J. Cameron and P. Cara, *Independent generating sets and geometries for symmetric groups*, J. Algebra 258 (2002), 641–650.

[3] J. Whiston, *Maximal independent generating sets of the symmetric group*, J. Algebra 232 (2000), 255–268.

[4] A. Maróti, *On the orders of primitive groups*, J. Algebra 258 (2002), 631–640, Theorem 1.1 in its original form.

[5] M. Conder, *The smallest regular polytopes of given rank*, Adv. Math. 236 (2013), 92–110. The nondegenerate rank-at-least-nine lower bound is retained as stated in the text; its full proof is not supplied here.